

Brian Caple

Senior Software Engineer & **Lead** Developer

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Summary

Senior Software Engineer & Lead Software Developer with 10+ years of experience in building scalable, high-performance applications across a variety of industries, including financial services, e-commerce, and SaaS. Proven expertise in designing and implementing cloud-native solutions, microservices architectures, and secure, real-time systems using .NET, ASP.NET Core, and related technologies. Skilled in DevOps practices, CI/CD pipelines, and infrastructure automation with Azure and AWS. Adept at optimizing system performance, ensuring security compliance, and leading teams in the development of innovative, enterprise-level applications. Strong focus on leveraging cutting-edge technologies such as event sourcing, CQRS, and containerization to solve complex business challenges.

Education

Bachelor's degree in **Computer and information science**
University of Texas | 2012

Visa Inc.

May 2021 – Dec 2024

Senior Software Engineer & Lead Developer

Austin, TX

As a **Senior Software Engineer & Lead Software Developer** at Visa Inc., I led the design and development of high-performance, scalable microservices-based solutions for a variety of global financial and supply chain platforms.

- Implemented **RESTful APIs** using **ASP.NET Core** for a global supply chain management platform, improving data retrieval speeds by optimizing database access patterns with **EF Core** and integrating JWT-based authentication for secure user access, resulting in improvement in API response times.
- Migrated a legacy on-premise ERP system to a cloud-based SaaS solution using **.NET 6**, implementing scalable **microservices** with **ASP.NET Core** and using **EF Core** for handling complex data models, reducing infrastructure costs while enhancing platform availability and scalability.
- Implemented an **API Gateway** using **Ocelot** in a **.NET Core**-based **microservices** architecture, consolidating requests to multiple backend services into a single entry point, reducing client-side complexity, and improving the scalability of a cloud-native application.
- Developed and deployed a high-availability invoicing platform using **ASP.NET Core** and **EF Core**, implementing features like multi-tenancy support and transactional email notifications, ensuring a seamless customer experience with real-time data updates and background tasks.
- Implemented Domain-Driven Design (DDD) in a multi-module **.NET** application to create a robust and scalable architecture for complex business logic, enhancing maintainability and aligning with business requirements across multiple domains like User Management, Billing, and Transactions.
- Implemented Command Query Responsibility Segregation (CQRS) in a **.NET-based SaaS application** to efficiently handle high transaction volumes by separating read and write models, reducing latency in query responses, and ensuring faster, more responsive user experiences.
- Implemented Event Sourcing in an enterprise-grade application using **.NET Core** to track state changes over time by persisting events rather than the current state, providing auditability, and enabling easy data recovery.
- Implemented event-driven workflows with **Node.js** and **RabbitMQ** for asynchronous task processing in an enterprise logistics platform, ensuring real-time data synchronization between distributed services and improving message throughput.
- Developed and optimized an enterprise-level payment gateway integration using **Node.js**, ensuring compliance with PCI-DSS standards. The integration reduced transaction processing times ensuring seamless payment experiences across different channels.
- Implemented a microservices architecture in **.NET** for an **e-commerce platform**, utilizing **SQL Server** for transactional data storage and **MongoDB** for handling unstructured product data, improving system scalability and performance.
- **Containerized a .NET Core microservices-based payment processing system** using **Docker**, enabling consistent environments across staging and production, streamlining the CI/CD pipeline, reducing deployment times, and increasing application reliability.
- Integrated **Azure Active Directory (AAD) authentication** for a SaaS platform, implementing **Single Sign-On (SSO)** across multiple applications, streamlining the user authentication process, resulting in increase in user adoption and reduction in support tickets related to login issues.
- Implemented centralized logging using **Serilog** for a **.NET-based ERP system**, integrating it with **Azure Application Insights** to monitor and trace business-critical transactions, ensuring quick identification and resolution of performance bottlenecks.
- Created a centralized logging system for a multi-component SaaS application using **.NET 7**, allowing real-time monitoring and better insights into system behavior.
- Integrated **Azure Redis Cache** into a **.NET Core** web application to enable distributed caching, significantly improving data retrieval times and reducing the load on **SQL Server** for frequently accessed data, such as user profiles and product catalogs.

- Implemented Infrastructure as Code (**IaC**) using **Azure Resource Manager (ARM)** templates to automate the deployment of Azure services for a global .NET-based customer portal, reducing manual configuration errors and ensuring consistent deployments.
- Designed and deployed a distributed caching solution using **Redis** and a multi-region **Azure** setup, reducing cross-region latency and providing fast, consistent access to session data for users across North America and Europe.
- Implemented secure user authentication and authorization using **ASP.NET Core Identity** and **EF Core**, ensuring role-based access control for an internal HR management system, enforcing strict security policies like MFA and token-based authentication.
- Implemented two-factor authentication (**2FA**) using **TOTP** (Time-based One-Time Password) for user logins in a financial services platform, increasing the security of user accounts.
- Implemented HMAC (Hash-based Message Authentication Code) for secure API request validation in a **.NET Core** e-commerce platform, ensuring data integrity and preventing replay attacks.
- Developed and deployed **secure gRPC APIs** with mutual **TLS** (mTLS) authentication between microservices in a high-traffic logistics application, ensuring encrypted and authenticated communication between services.
- Created a **real-time data processing service** using **ASP.NET Core** and **SignalR**, providing live updates on stock market trends, integrated with **EF Core** for data storage and ensuring scalability through **Azure Kubernetes Service (AKS)**.
- Used **NServiceBus** and **ActiveMQ** to implement reliable, high-throughput messaging queues, ensuring critical business operations such as payment processing and inventory updates were executed with minimal latency in a .NET-based distributed system.
- Implemented an **Angular-based front-end** for a cross-platform enterprise application in a **.NET** environment, leveraging Angular's modularity to streamline development and enhance maintainability, while integrating data visualizations and interactive dashboards for an e-commerce platform.
- Designed and integrated secure, role-based user authentication and authorization features in an **Angular** front-end, connecting to **ASP.NET Identity** and a **.NET Core** API backend, ensuring secure access control for enterprise users in a financial services application.
- Developed and maintained automated **end-to-end UI tests** using **Selenium WebDriver** for a .NET-based customer portal, ensuring consistent behavior across browsers and improving test coverage for user authentication, data entry, and report generation.
- Designed and executed integration tests using **xUnit** and **TestServer** to validate the interaction between the API layer and database in a .NET-based **SaaS solution**, ensuring correct data flow and accurate responses.
- Leveraged dependency injection in a **.NET Core** application to facilitate easier unit testing by injecting mock services (using **Moq**) for external dependencies like databases and third-party APIs.
- Refined the architecture of a large-scale enterprise HR management system by implementing a **layered** architecture in **.NET 7**, improving maintainability, scalability, and reducing onboarding time for new developers.
- Implemented **in-memory caching** using **.NET MemoryCache** for frequently accessed API responses, reducing server load and improving response time in a retail enterprise system.
- Implemented centralized logging using **Log4Net** for a **.NET-based ERP application** to track transaction logs for order processing, inventory management, and financial transactions, improving auditability and compliance with data regulations.
- Created a set of SOAP Web Services in **.NET Framework** for a legacy supply chain management system, enabling secure communication with external partners while ensuring compatibility with existing business logic and data models.
- Adopted **Agile methodologies (Scrum)** for a .NET-based enterprise application, increasing delivery speed by iterating in 2-week sprints, organizing daily standups, and maintaining clear sprint backlogs.
- Facilitated **CI/CD pipeline setup** using **GitHub Actions** for a **.NET-based SaaS solution**, automating the build, test, and deployment processes for faster iterations.
- Integrated **real-time data analytics** into a **.NET** SaaS platform using **Azure Analytics** services, feeding insights into **Power BI** dashboards, providing visibility into metrics like user activity and system performance.

- Optimized **complex T-SQL queries in a large-scale enterprise application** by identifying and indexing frequently queried tables, resulting in a improvement in query performance and reducing database load during peak usage.
- Developed stored procedures and functions in **T-SQL** to automate data processing tasks, reducing manual intervention and ensuring consistent and accurate data transformations for reporting and analytics.
- Implemented transactional support with **T-SQL in a .NET-based enterprise application**, ensuring ACID compliance for critical business processes, preventing data inconsistencies and ensuring high reliability in data operations.
- Architected and implemented **Azure App Services** for hosting a .NET-based enterprise application, streamlining deployment processes with automated scaling and reducing infrastructure management overhead.
- Utilized **Azure Key Vault** for managing sensitive application secrets (such as API keys and connection strings), ensuring secure storage and retrieval of credentials, and complying with industry best practices for security and compliance.
- Implemented Azure Active Directory (Azure AD) for Single Sign-On (SSO) in a multi-tenant SaaS enterprise application, enhancing user security and simplifying access management for both internal and external users.

Holman

Nov 2020 – Jan 2021

Senior Software Engineer

Grapevine, TX

As a **Senior Software Engineer** at Holman, I led the design and implementation of a modular, scalable microservices architecture for large-scale enterprise applications, leveraging .NET Core and Azure technologies.

- Designed and implemented a **Microservices Architecture** for a large-scale enterprise application using .NET Core, enabling modular development and independent deployment cycles. Each microservice was responsible for a specific domain, such as payment processing, notifications, and user authentication, and communicated via **REST APIs**, ensuring high scalability and fault tolerance.
- Built a fully decentralized **microservices-based platform** in .NET for a global retail enterprise, allowing the organization to scale independently by deploying services like inventory management, order processing, and shipping separately. This modular approach reduced deployment risk and enabled the team to innovate faster while maintaining consistency through **API gateways** and event-driven communication.
- Designed and implemented a **Service-Oriented Architecture (SOA)** for a financial application in .NET, enabling seamless integration between various legacy systems and new services through SOAP and **RESTful APIs**, improving modularity and scalability across distributed systems.
- Developed and implemented end-to-end **encryption** for communication between mobile applications and backend servers, using **SSL/TLS certificates** to secure data in transit and protect user privacy, ensuring compliance with GDPR for personal data protection.
- Built automated **data migration tools** using **EF Core** for synchronizing legacy database schemas with modern cloud-based databases, facilitating seamless migration with minimal downtime and ensuring data integrity across systems in a large-scale enterprise application.
- Developed robust **SQL Server-based data models** for enterprise-level applications, optimizing queries and stored procedures to handle high-volume transactions, ensuring data consistency and integrity in financial reporting systems, and supporting real-time analytics.
- Developed and optimized **RESTful APIs** using **ASP.NET Core** for an enterprise e-commerce platform, enabling seamless integration with external payment gateways, reducing transaction processing time and supporting over 500,000 concurrent users.
- Designed and implemented a **GraphQL API** layer using **HotChocolate** in .NET Core, allowing clients to query only the necessary data for a product recommendation engine, reducing data transfer size and improving client-side performance for a retail app.

- Implemented comprehensive **unit tests** using **MSTest** and **Moq** to validate core business logic in a .NET Core-based financial application, ensuring the accuracy and reliability of calculations and data transformations before integration into the full system.
- Established a **DevOps culture** by integrating **Azure DevOps** with the development pipeline for a large-scale .NET Core eCommerce platform. This automated code builds, testing, and deployment to multiple environments, reducing manual intervention, speeding up delivery cycles, and enhancing the overall quality of releases.

Technamo LLC

Oct 2015 – Aug 2020

Senior Software Developer

Austin, TX

As a **Senior Software Developer** at Technamo LLC, I played a key role in architecting and deploying scalable, high-performance microservices for enterprise applications, utilizing .NET Core, AWS, and Azure technologies.

- Implemented **Infrastructure as Code** (IaC) with **Terraform** for a .NET-based microservices application deployed across **AWS** and **Azure**, enabling the automation of infrastructure provisioning, scaling policies, and resource management. This approach allowed the DevOps team to provision environments quickly and reliably, reducing deployment times.
- Architected and implemented an **AWS S3-based** storage solution for a cloud-native enterprise application, enabling secure and scalable file storage with automated backup and retrieval, reducing operational overhead.
- Configured **Azure Load Balancer** and **Application Gateway** to distribute traffic across multiple instances of a .NET Core web application, ensuring high availability and minimizing downtime during peak loads for a global financial services platform.
- Designed and implemented a horizontally scalable **microservices** architecture in .NET Core using **Docker** and **Kubernetes**, ensuring seamless load balancing and high availability across multiple instances of the services for a real-time event processing application.
- Implemented horizontal scaling strategies for **NoSQL** databases like **Cassandra** in a high-availability environment, supporting the growth of user data and ensuring that the application remained performant even under heavy loads.
- Optimized database schemas for high-performance transaction processing in an enterprise banking application, reducing query execution time through advanced **indexing** strategies and normalization techniques, ensuring high availability and responsiveness.
- Designed and optimized a data warehousing solution using **SQL Server** and **Azure Synapse Analytics** for an enterprise analytics platform, enabling near real-time data aggregation and providing actionable insights for business intelligence reports.
- Created a multi-tier data replication architecture between on-premises **SQL Server** and **Azure SQL Database** using transactional replication, improving the reliability of critical business processes and reducing downtime during system migrations.
- Streamlined data access and improved application performance by refactoring **EF Core** queries in an **e-commerce** platform to utilize efficient **LINQ-to-SQL** translations, reducing server load and improving response times for inventory searches.
- Developed and deployed **gRPC-based** microservices using .NET Core for an internal order management system, enabling high-performance communication between services with minimal latency, resulting in a improvement in inter-service communication speed compared to **RESTful APIs**.
- Developed and maintained **Entity Framework (EF)** Core data models for an **inventory management system**, reducing the complexity of data access layers and improving the maintainability of the application by leveraging EF's Code-First approach for rapid development and migrations.
- Designed and implemented **OAuth 2.0** and **OpenID Connect** protocols for secure, token-based user authentication in a .NET Core web application, enabling seamless integration with third-party services while ensuring compliance with modern security standards.

- Developed a custom identity management solution using **ASP.NET Identity**, enabling role-based authorization and fine-grained access control for a healthcare application, ensuring sensitive data was only accessible by authorized personnel, and meeting regulatory requirements such as **HIPAA**.
- Utilized **IdentityServer4** in a .NET Core application to implement centralized identity management for a multi-tenant **SaaS** application, enabling secure and consistent user authentication and authorization across all tenants while supporting social logins and external identity providers.
- Created a secure **SOAP Web Service** in .NET to handle sensitive financial data transactions for a global banking application, implementing **WS-Security** for encrypted communication and ensuring compliance with industry regulations such as **PCI DSS**.
- Refactored critical application services using **asynchronous programming** patterns (**async/await**) in .NET, improving scalability and reducing the response time for I/O-bound operations, such as database queries and external API calls, in a high-traffic **SaaS** platform.
- Implemented message queues to facilitate asynchronous communication between microservices in a .NET-based enterprise application, reducing tight coupling and enhancing system scalability and fault tolerance.
- Designed and implemented event-driven workflows to enable real-time notifications and data synchronization across distributed systems in a large-scale **SaaS** platform, leveraging **Azure Event Grid** and .NET for seamless event publishing and consumption.
- Designed and implemented an event-driven architecture using **gRPC** for inter-service communication in a cloud-based supply chain management system, improving scalability and reducing network traffic by enabling direct, low-latency communication between microservices.
- Implemented **Azure Monitor** and **Application Insights** for a .NET-based microservices application, setting up real-time monitoring of application performance, server health, and user interactions. This integration enabled proactive issue detection, allowing the team to resolve performance bottlenecks and application errors quickly, reducing downtime.
- Developed custom data pipelines to feed real-time data from a .NET Core application into **Power BI** for visualizing customer engagement and financial transactions. This helped the marketing and finance teams gain insights into customer behavior patterns and optimize marketing campaigns, resulting in increase in customer retention.
- Integrated **Power BI** with a .NET Core application to provide real-time reporting and data visualization capabilities to senior management. This allowed users to interactively view business metrics such as sales performance, customer behavior, and operational KPIs, leading to more informed decision-making and an improvement in operational efficiency.
- Conducted regular penetration testing and vulnerability assessments using tools such as **OWASP ZAP** and **Burp Suite** to identify and mitigate potential security flaws in a cloud-based supply chain management platform, resulting in the resolution of critical vulnerabilities before production deployment.

Leobit

Jan 2013 – Aug 2015

Software Developer

Austin, TX

- Integrated **Redis** as a distributed caching layer in a .NET-based e-commerce application, reducing database load and improving response times for frequently accessed data like product details and user sessions.
- Created a secure API layer for an e-commerce platform by integrating **JWT** (JSON Web Tokens) authentication, implementing token-based user sessions and providing role-based access control (RBAC) to ensure that only authorized users could access sensitive product and order data.
- Implemented **API rate-limiting** and **IP-based security** policies in an API Gateway for a financial application, ensuring compliance with regulatory requirements and preventing abuse of critical financial endpoints, while improving overall API security.

- Implemented **AES-256 encryption** for sensitive customer data in a financial application using .NET Data Protection APIs, ensuring compliance with PCI DSS and reducing the risk of data breaches during both transmission and storage.
- Optimized complex stored procedures and **EF queries** for a large-scale logistics application, improving data retrieval times and ensuring the efficiency of order tracking and shipment updates in real-time.
- Implemented **Test-Driven Development (TDD)** for a .NET Core-based eCommerce platform, writing unit and integration tests using xUnit and Moq to ensure robust order processing, payment gateway integration, and inventory management features. This led to a reduction in defects during the development cycle and improved collaboration between development and QA teams.
- Containerized a .NET Core microservices-based payment processing system using **Docker**, enabling developers to work in consistent environments and simplifying the deployment process across various environments (staging, production). This streamlined the **CI/CD pipeline**, reducing deployment times and increasing the overall reliability of the application.
- Implemented **Docker Compose** for local development environments in a .NET microservices architecture, enabling development teams to simulate production-like environments with ease. This eliminated discrepancies between development and production environments, reducing deployment issues.
- Designed and implemented a **hybrid cloud solution** for a global SaaS provider using **Azure** and **AWS** to balance workloads between on-premise servers and public cloud instances. This solution allowed the company to optimize resource allocation, reduce costs, and improve disaster recovery capabilities for their .NET-based application, ensuring seamless failover and scaling during high-demand periods.

Technical skills

- **API Development & Integration:** RESTful APIs, SOAP, GraphQL, gRPC, API Gateway (Ocelot), API Security, Rate-Limiting, JWT, OAuth 2.0, API Management
- **Authentication & Authorization:** ASP.NET Core Identity, JWT, OAuth 2.0, OpenID Connect, Single Sign-On (SSO), Multi-Factor Authentication (MFA), Role-Based Access Control (RBAC), HMAC
- **Microservices & Architecture:** Microservices Architecture, Event-Driven Architecture, Domain-Driven Design (DDD), Command Query Responsibility Segregation (CQRS), Event Sourcing, Service-Oriented Architecture (SOA), gRPC, API Gateway, Azure Service Bus, RabbitMQ, NServiceBus
- **Cloud Platforms & Services:** AWS (EC2, RDS, S3, CloudWatch), Azure (App Services, Azure Active Directory, Key Vault, Redis Cache, Application Insights, Azure Event Grid), AWS CloudFormation, Azure Resource Manager (ARM) templates
- **Infrastructure & DevOps:** Infrastructure as Code (IaC), Terraform, Docker, Kubernetes, CI/CD Pipelines (Azure DevOps, GitHub Actions), Automated Deployments, Containerization, Docker Compose
- **Database & Data Management:** SQL Server, NoSQL (Cassandra, MongoDB), Azure SQL Database, Entity Framework (EF) Core, T-SQL, Stored Procedures, Data Caching (Redis), SQL Query Optimization, Data Replication, Data Warehousing
- **Performance & Scalability:** Horizontal Scaling, Load Balancing (Azure Load Balancer, Application Gateway), Distributed Caching, Redis, Query Optimization, High-Availability Solutions, Asynchronous Programming, EF Core Performance Tuning
- **Security & Compliance:** End-to-End Encryption (SSL/TLS), AES Encryption, HMAC, PCI-DSS Compliance, WS-Security, GDPR, Secure APIs, Token-Based Authentication, Secure gRPC Communication (mTLS)
- **Real-Time Systems & Messaging:** SignalR, WebSockets, Real-Time Data Processing, Event-Driven Workflows, Message Queues (AWS SNS, SQS, ActiveMQ), Distributed Messaging
- **Logging & Monitoring:** Centralized Logging (Serilog, Log4Net), Application Insights, Azure Monitor, CloudWatch, Real-Time Monitoring, Performance Bottleneck Identification, Automated Alerting
- **Front-End Development:** Angular, SPA Development, Data Visualization, Interactive Dashboards, UI Optimization
- **Testing & Quality Assurance:** Unit Testing (xUnit, MSTest, Moq), Test-Driven Development (TDD), Integration Testing, End-to-End UI Testing (Selenium WebDriver), Performance Testing
- **Data Analytics & Reporting:** Power BI Integration, Real-Time Analytics, Data Pipelines, Azure Synapse Analytics, Data Visualization
- **Development Methodologies:** Agile (Scrum), DevOps, Continuous Integration/Continuous Deployment (CI/CD), Automated Testing, Version Control (Git, GitHub, Azure DevOps)